SW/CV Challenge

Build an expressive 6-DOF lamp system that sees, reacts, remembers, and converses. This challenge tests your ability to design a full perception-to-action pipeline — the core technical problem behind LeLamp.

System Overview

Your lamp should behave like a social, aware agent:

1. **Understands** when someone is paying attention
2. **Reacts** with expressive behaviors using motion, light, and sound
3. **Remembers** what it has seen
4. **Recalls** past events conversationally via LLM

Demo Scenario

Step	What happens
1. Engagement Detection	User looks at the lamp → interaction begins. User looks away → lamp disengages.
2. Attention-Seeking Behavior	User stays disengaged → lamp attempts subtle re-engagement (movement, light, sound).
3. Memory Formation	Lamp observes the scene — identifies and classifies objects in view.
4. Memory Recall	User later asks about an object's location → lamp correctly answers using its stored memory.

Deliverables

1. Technical Writeup

- System architecture diagram
- Data flow (high-level and low-level)
- Design decisions and tradeoffs

2. Demo Video

- Working demonstration of the four steps above

3. Evaluation

- Engagement detection reliability metrics
- End-to-end latency measurements

What We're Looking For

Criteria	What matters
Architecture	Is the system well-structured? Are the modules cleanly separated?
Perception pipeline	Does engagement detection work reliably in real-time?
Behavior design	Are the lamp's reactions expressive and natural?
Memory system	Can the lamp store and retrieve scene information accurately?
Tradeoff reasoning	Have you explained *why* you chose this approach over alternatives?
Code quality	Is the code clean, modular, and well-documented?

Bonus Challenges *(optional, for standout candidates)*

- **Multi-user interaction** — identify who is speaking
- **Emotion detection** from voice tone
- **Self-learning behavior** that improves over time
- **Interruption awareness** — knows not to interrupt when the user is talking or watching TV